

TrES-1b

R-1926

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_130815 · created 2026-08-06T01:14:42+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456519.883 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.053
Transit depth (Rp/Rs)^2	1.6e-05 +/- 0.0042 [%]
Orbital Inclination (inc)	85.18 +/- 1.43
Airmass coefficient 1 (a1)	0.97 +/- 0.64
Airmass coefficient 2 (a2)	0.002 +/- 0.083
Scatter in the residuals of the lightcurve fit is	64.11 %
Best Comparison Star	#3 - [551, 426]
Optimal Aperture	0.87
Optimal Annulus	5.75
Transit Duration (day)	0.041 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0 ± 0.053 vs 0.1358 (deviation 2.11σ)
Duration fit vs published	0.041 d vs 0.1042 d (deviation 1.68σ)
Mid-time O–C	-1.8 min
Depth SNR	0.0
Residual scatter	64.11 %

Light curve

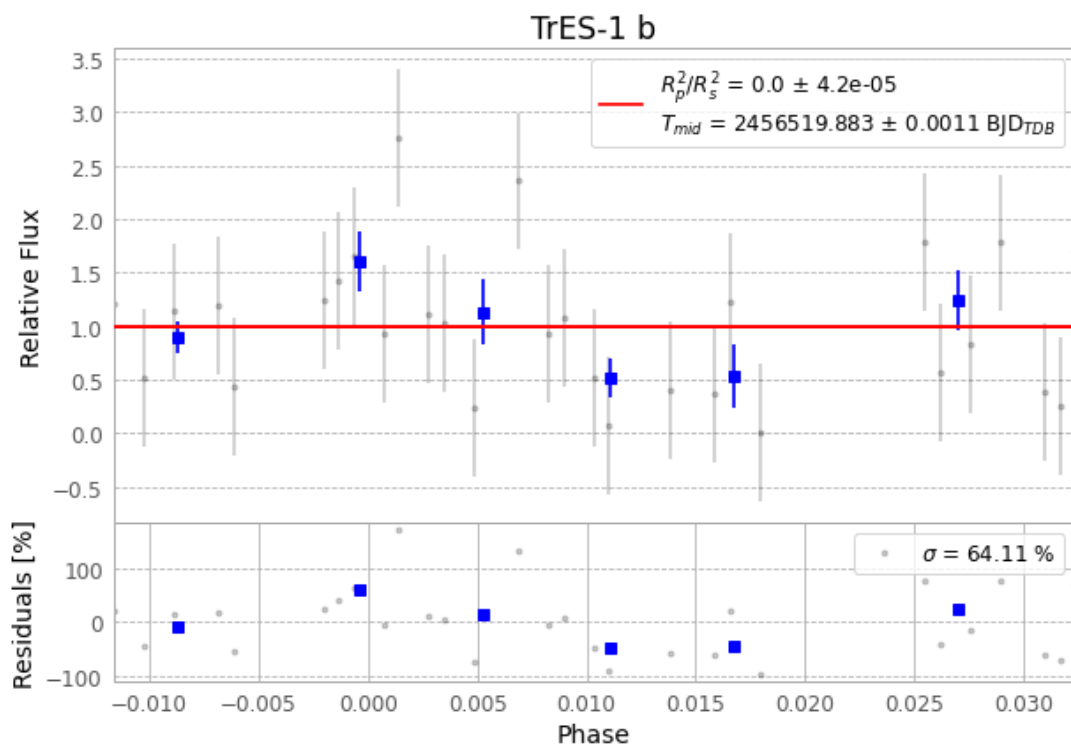


Figure 1 — FinalLightCurve_TrES-1 b_2013-08-15.png (SHA-256 7bde5eebe65e02d7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [194, 290], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 4958ce1586467d8a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

66 FITS frames (43 MB), TRES-1130815081213.FITS → TRES-1130815113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-130815.log (ed52a6d0fde1...), FinalLightCurve_TrES-1 b_2013-08-15.png (7bde5eebe65e...), FinalLightCurve_TrES-1 b_2013-08-15.csv (472f4adf52e1...)

Compute resources

Wall time 111.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 01:14Z.